

ESTIAK AHMED

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RESEARCH PROFILE

- Transportation infrastructure and pavement management, with emphasis on UAV-based condition assessment, AI-assisted infrastructure inspection, and transportation asset management.
- Computer vision, machine learning, geospatial analytics, and remote sensing for civil infrastructure applications, including YOLO/CNN workflows, SfM photogrammetry, and Web GIS.
- BIM and smart infrastructure applications including digital twins, 4D simulation, fire safety modeling, evacuation analysis, and human-building interaction.

EDUCATION

Master of Science in Civil Engineering

University of Wyoming, Laramie, WY, USA

Aug 2025 - Present

- CGPA: 3.83 / 4.00
- Department: Civil & Architectural Engineering and Construction Management
- Thesis Supervisor: Ahmed Abdelaty, Ph.D.

Selected Graduate Coursework: Pavement Management System, AI in Transportation, Machine Learning & Data Analysis, UAV Data Acquisition and Processing, Spatial Data Science, Web Mapping and Internet GIS, Remote Sensing.

Bachelor of Science in Building Engineering & Construction Management

(BECM)

Khulna University of Engineering & Technology (KUET), Bangladesh

Nov 2018 - Mar 2024

- CGPA: 3.39 / 4.00
- Undergraduate Thesis: Development of a BIM-Based Fire Safety Simulation Framework and Evacuation Planning
- Supervisor: Md. Abu Safayet

RESEARCH EXPERIENCE

Graduate Research Project - UAV-Based Pavement Crack Detection

University of Wyoming | Advisor: Ahmed Abdelaty, Ph.D.

2026 - Present

- Investigating computer-vision-based pavement crack detection using UAV imagery for pavement condition assessment and management.
- Applying UAV data acquisition, photogrammetric processing, geospatial analysis, and machine learning workflows to evaluate pavement distress in a Wyoming case study.
- Comparing computer vision model performance and translating detection results into pavement-management-relevant findings.
- Prepared a first-author manuscript recommended for presentation at the Transportation Research Board (TRB) Annual Meeting 2027.

Undergraduate Research - BIM-Based Fire Safety and Evacuation Simulation

Khulna University of Engineering & Technology (KUET), Bangladesh

2023 - 2024

- Developed a BIM-based framework for analyzing building fire safety and evacuation planning using computer simulation.
- Integrated building information modeling with fire/smoke and pedestrian evacuation analysis to evaluate emergency scenarios without relying on full-scale evacuation testing.
- Applied simulation-oriented workflows relevant to life safety, occupant movement, and built-environment performance.

PUBLICATIONS & SCHOLARLY ACTIVITY

Peer-Reviewed Journal Article

- Anzum, A., Rana, M. J., **Ahmed, E.**, & Rahman, S. (2025). Determining the optimal window size and orientation of an academic building in subtropical climate. *International Journal of Building Pathology and Adaptation*. doi:10.1108/IJBPA-11-2024-0232

Peer-Reviewed Conference Proceedings & Presentations

- Joy, M. K., **Ahmed, E.**, & Anzum, A. (2026). A comprehensive review and paradigm framework on Human-AI collaboration in construction management. Presented at the 1st International Conference on Industrialized

Construction (ICoNIC 2026), University of Florida, Gainesville, FL.

- **Ahmed, E.**, Bhattacharjee, A., & Anzum, A. (2025). Reducing false alarms in buildings with UAV-based fire detection and monitoring through a cloud-integrated approach. *Advances in Engineering Research*, 120-127. doi:10.2991/978-94-6463-884-4_15
- Bhattacharjee, A., **Ahmed, E.**, & Rana, M. J. (2025). Automation in construction quality control: Integrating Unmanned Aerial Vehicle (UAV), CNNs, and photogrammetry for structural element detection and measurement precision. *Advances in Engineering Research*, 181-189. doi:10.2991/978-94-6463-884-4_22

Recommended for Presentation

- **Ahmed, E.**, Abdelaty, A., & Khan, S. Comparing computer vision based models for UAV-based pavement crack detection in pavement management: A case study in Wyoming. Recommended for presentation at the Transportation Research Board (TRB) Annual Meeting, Washington, D.C., 2027.

PEER REVIEW ACTIVITY

- Reviewer, *Journal of Building Engineering* (Elsevier)
- Reviewer, *International Journal of Building Pathology and Adaptation* (Emerald Publishing)

ACADEMIC & TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug 2025 - Present

Department of Civil & Architectural Engineering and Construction Management, University of Wyoming

HVAC of Buildings (ARE 3400 / ME 3400) | Fall 2026 - Present

- Conduct class sessions and support instruction in moist-air properties, psychrometrics, air-conditioning processes, comfort and health, indoor environmental quality, and heat transmission through building structures.
- Develop homework assignments, answer keys, instructional materials, and course content to support lectures and student learning.
- Grade homework assignments, evaluate student work, and provide feedback aligned with course learning objectives.

Engineering Surveying (CE 2070) | Aug 2025 - Spring 2026

- Served as primary lab and field instructor for 2+ lab sections totaling 40+ students, leading total-station positioning, leveling, topographic mapping, and field surveying practicums.
- Designed and revised lab exercises and grading rubrics; guided students through spatial data collection and measurement error correction.

Exploring CAECM (ARE/CE/CM 1000) | Aug 2025 - Dec 2025

- Graded and assessed approximately 120 first-year students across Civil Engineering, Architectural Engineering, and Construction Management.

SELECTED TECHNICAL SKILLS

Category	Software / Methods
AI / Computer Vision	Python, NumPy, scikit-learn, PyTorch, YOLO/CNN workflows
UAV / Photogrammetry / Point Clouds	Pix4D, Meshroom, CloudCompare, structure-from-motion workflows
GIS / Remote Sensing	QGIS, ArcGIS Desktop, ERDAS Imagine, Web GIS, JavaScript
BIM / Digital Construction	Autodesk Revit, Navisworks Manage, AutoCAD, Design Review
Simulation	PyroSim, Pathfinder, ETABS, Robot Structural Analysis
Data Analysis / Programming	Python, R, JavaScript, C, Fortran
Additional Tools	SketchUp, Enscape, Lumion, Adobe Illustrator, Photoshop, Premiere Pro, Camtasia, Canva

CERTIFICATIONS & LICENSES

- FAA Remote Pilot Certificate (Part 107), Federal Aviation Administration
- International Teaching Assistant (ITA) Training Certificate, University of Wyoming
- Supervised Machine Learning: Regression and Classification, DeepLearning.AI / Stanford University, Nov 2024
- Construction Project Management, Columbia University, Oct 2024
- The Effect of Fires on People, Property and the Environment, University of Maryland, College Park, Oct 2024
- Google IT Support Professional Certificate, Google, Jul 2021

HONORS & LEADERSHIP

- Vice President, UW ITE (Institute of Transportation Engineers) Student Chapter, Apr 2026 - Present
- Winner, CAD Master of Civil Convivial 3.0
- Vice President, BUILTECH FEST 5.0, Sep 2023 - May 2024
- Vice President, BECM Technical Club (BTECK), KUET, May 2023 - May 2024
- Class Representative, Department of BECM, KUET, Feb 2020 - Sep 2023

PROFESSIONAL MEMBERSHIP

- Associate Member (AMIEB), Institution of Engineers, Bangladesh (IEB), ID: 27574

SELECTED RESEARCH & TECHNICAL PROJECTS

Automated Road Surface Detection from USGS 3DEP LiDAR

GIST 5450 course project

- Developed a two-stage road-surface extraction workflow using USGS 3DEP LiDAR, CloudCompare/CANUPO, Random Forest, DBSCAN, and PCA-based morphological filtering.
- Processed a 10.35-million-point I-80 tile and applied the trained model to an independent 5.50-million-point US-287 tile; LiDAR intensity was the dominant predictor (68.1% feature importance).
- Reduced 361 predicted spatial clusters to 8 elongated road candidates through DBSCAN clustering and PCA filtering, while documenting limitations including rail-line confusion and road loss at curves and intersections.

Comparative Performance Verification of Multi-Generational YOLO Architectures for Building Crack Detection

ATSC 5009 course project

- Benchmarked YOLOv8, YOLOv11, YOLOv12, and YOLO26 on the METU Concrete Crack Dataset under identical training conditions using precision, recall, mAP@50, and inference speed.
- YOLOv11 achieved the highest precision (93.2%) and mAP@50 (0.896), while YOLOv12 achieved the highest recall (81.9%) and YOLOv8 provided the fastest inference (2.50 ms).

Coastal Land-Water Change Detection in Bangladesh Using Landsat and NDWI

GIST 5111 course project

- Analyzed Landsat imagery from 2015 and 2025 using the Normalized Difference Water Index (NDWI) to classify water and non-water areas in coastal Bangladesh.
- Quantified an 11.5% decrease in non-water area and a 15.4% increase in water area over the study period, demonstrating remote-sensing change detection for coastal landscape assessment.

Interactive Construction Site Monitoring WebGIS Application

GIST 5300 course project

- Developed a custom WebGIS application using the ArcGIS JavaScript API, ES modules, GeoJSON, and custom CSS, with project-type filtering, heatmaps, popups, click-to-zoom functionality, and layer controls.

UAV-Based Fire Detection and Cloud-Integrated Monitoring

Peer-reviewed conference research

- Investigated UAV-enabled fire detection and monitoring with a cloud-integrated workflow aimed at reducing false alarms in building applications.
- Published the work in *Advances in Engineering Research* in 2025.

Automated Construction Quality Control with UAV, CNNs, and Photogrammetry

Peer-reviewed conference research

- Studied integration of UAV imagery, convolutional neural networks, and photogrammetry for structural element detection and measurement-oriented construction quality control.
- Published the work in *Advances in Engineering Research* in 2025.

Human-AI Collaboration in Construction Management

Conference presentation

- Contributed to a comprehensive review and paradigm framework examining Human-AI collaboration in construction management, presented at IConIC 2026 at the University of Florida.

ADDITIONAL EXPERIENCE

Transaction Service Officer (Trainee), IFIC Bank PLC, Bangladesh

Jun 2025 - Jul 2025

- Supported customer account operations, fund transfers, and digital banking services within a One-Stop Banking model.